

Subject Description Form

Subject Code	ITC6854RT																																												
Subject Title	Guided Study in Textile Structure and Mechanics																																												
Credit Value	3																																												
Level	6																																												
Pre-requisite / Co-requisite / Exclusion	Nil																																												
Objectives	To understand fundamental concepts, theories and models on textile/fabric structure and mechanics. To apply the classic models and emerging approaches to structure and mechanics of textile/fabric materials in different projects.																																												
Intended Learning Outcomes	Upon completion of the subject, students will be able to: - Apply different concepts, theories and models on textile/fabric structure and mechanics - Catch the up-to-date information of structure and mechanics of textile materials - Understand different approaches used in textile/fabric mechanics analysis																																												
Subject Synopsis/ Indicative Syllabus	This subject will provide foundation and guide for research students to carry out in-depth study of textile/fabric structure and mechanics in terms of concepts and theories, models, applications and current status as well as future development. In addition, general approaches used in textile/fabric mechanics analysis such as energy methods, force methods, numerical mechanics/computational mechanics will be dealt with.																																												
Teaching/Learning Methodology	This is a guided study course and will have derived self-studying. The subject lecturer will suggest reading materials based on the student's research area and discuss with the students. Students are requested to give oral presentations and prepare written reports based on the reading materials.																																												
Assessment Methods in Alignment with Intended Learning Outcomes	<table border="1" style="width: 100%; border-collapse: collapse;"> <thead> <tr> <th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="6">Intended subject learning outcomes to be assessed</th></tr> <tr> <th>a</th><th>b</th><th>c</th><th></th><th></th><th></th></tr> </thead> <tbody> <tr> <td>Continuous assessment</td><td>40%</td><td style="text-align: center;">✓</td><td style="text-align: center;">✓</td><td style="text-align: center;">✓</td><td></td><td></td><td></td></tr> <tr> <td>Examination</td><td>60%</td><td style="text-align: center;">✓</td><td style="text-align: center;">✓</td><td style="text-align: center;">✓</td><td></td><td></td><td></td></tr> <tr> <td>Total</td><td>100%</td><td colspan="6"></td></tr> </tbody> </table> Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Continuous assessment will be used to train students' knowledge and skills learned in this subject and examination will be used to assess the overall learning outcomes of students.							Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed						a	b	c				Continuous assessment	40%	✓	✓	✓				Examination	60%	✓	✓	✓				Total	100%						
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Student Study Effort Expected	Class contact: ▪ Tutorial and discussion					26 Hrs.																																							

	Other student study effort:	
	▪ Reading	56 Hrs.
	▪ Preparation of reports	44 Hrs.
	Total student study effort	126 Hrs.
Reading List and References	1. Schwartz, Peter, ed. Structure and mechanics of textile fibre assemblies. Woodhead publishing, 2019. 2. Behera, Bijoy Kumar, and P. K. Hari. Woven textile structure: Theory and applications. Elsevier, 2010. 3. Hu, Jinlian. Structure and mechanics of woven fabrics. Elsevier, 2004. 4. Zulifqar, Adeel, Zubair Khaliq, and Hong Hu. "Textile Mechanics: Woven Fabrics." Handbook of Fibrous Materials (2020): 455-476. 5. Pan, Ning, and Phil Gibson, eds. Thermal and moisture transport in fibrous materials. Woodhead Publishing, 2006.	